Reviewer cWSM 3

Comment 3.1: Surgery appears to be highly sensitive to hyperparameters, which may make it difficult to adapt efficiently to models of different sizes or to new datasets without extensive tuning.

Response 3.1:

W1: Is Surgery highly sensitive to hyperparameters, making it difficult to adapt to different model sizes or new datasets without extensive tuning?

(Clarification on why a relatively large hyperparameter λ is required). As illustrated in Figure 2, the magnitude of the sink divergence d_h is generally small, with most attention heads having d_h values concentrated within the range $[-0.05, 0.05]$. After applying the ReLU function in Equation (4), these values are further compressed into the interval $[0, 0.05]$. As a result, the regularization term becomes numerically weak; without a sufficiently large hyperparameter λ, its contribution to gradient updates during backpropagation would be minimal. Therefore, a relatively large λ is necessary to amplify the effect of this regularization term and ensure it meaningfully influences the optimization process.

In this rebuttal, we address the reviewer's concerns by further evaluating the sensitivity of the hyperparameter λ across models of different sizes and diverse datasets. Please see the following results.

(With different model sizes s)

	Harmful Score ↓	**->**	**->**	**Finetune Accuracy ↑**	**->**	**->**
Methods	s=1B	s=8B	s=14B	s=1B	s=8B	s=14B
SFT	37.00	27.60	27.30	40.30	64.70	90.70
Surgery (λ = 0.1)	39.00	25.00	24.50	40.60	67.40	90.20
Surgery (λ = 10)	31.90	10.70	17.50	40.20	68.10	90.40
Surgery (λ = 100)	24.40	9.00	12.80	38.90	68.50	90.50
Surgery (λ = 300)	18.90	8.90	11.30	38.60	68.50	90.80
Surgery (λ = 500)	15.80	8.40	22.20	36.60	68.40	86.70

The above results show that λ **does not exhibit significant model dependency or high sensitivity across different model sizes, but instead shows strong robustness and transferability**. Specifically, Surgery consistently improves defense performance (i.e., decreasing HS) as λ increases across models of different scales (i.e., Llama-3.2-1B-Instruct, Llama-3-8B-Instruct, and Qwen2.5-14B). Meanwhile, for larger models, when λ lies within the range of 10–300, the inference performance is also improved. To achieve a balance between defense effectiveness and inference performance, $\lambda = 300$ demonstrates stable and superior performance across all three model scales.

(With different fine-tuning datasets)

	Harmful Score ↓	**->**	**->**	**Finetune Accuracy ↑**	**->**	**->**
Methods	SST2	GSM8K	AGNEWS	SST2	GSM8K	AGNEWS
SFT	34.20	27.60	31.00	94.50	64.70	90.30
Surgery (λ=0.1)	29.10	25.00	32.90	94.95	67.40	90.50
Surgery (λ=10)	14.30	10.70	18.20	94.50	68.10	90.10
Surgery (λ=100)	9.50	9.00	8.40	94.38	68.50	88.70
Surgery (λ=300)	9.40	8.90	7.40	94.50	68.50	88.60
Surgery (λ=500)	9.30	8.40	7.90	93.69	68.40	89.90

The results indicate that λ **does not exhibit significant dataset dependency, but instead demonstrates strong robustness and transferability**. Specifically, Surgery consistently improves defense performance (i.e., decreasing HS) as λ increases across different datasets (i.e., SST2, GSM8K, and AGNEWS). When λ is around 300, it achieves a good balance between defense effectiveness and inference performance across all three datasets.

(Summary) The above results demonstrate that **Surgery does not exhibit significant hyperparameter dependency or high sensitivity across models of different scales and diverse fine-tuning datasets**. A unified hyperparameter configuration can achieve stable and effective performance across various settings, indicating strong robustness and generalizability.

Comment 3.2: Surgery adopts a high regularizer intensity (=300), which raises significant concerns about potential over-refusal problem. It would be better to evaluate the fine-tuned model on over-refusal benchmarks like OKTest (https://arxiv.org/abs/2401.17633) and FalseReject (https://arxiv.org/abs/2505.08054).

Response 3.2:
W2: Does Surgery, which adopts a high regularization intensity (λ = 300), lead to an over-refusal problem?

(Clarification on why a relatively large hyperparameter λ is required). As illustrated in Figure 2, the magnitude of the sink divergence d_h is generally small, with most attention heads having d_h values concentrated within the range $[-0.05, 0.05]$. After applying the ReLU function in Equation (4), these values are further compressed into the interval $[0, 0.05]$. As a result, the regularization term becomes numerically weak; without a sufficiently large hyperparameter λ, its contribution to gradient updates during backpropagation would be minimal. Therefore, a relatively large λ is necessary to amplify the effect of this regularization term and ensure it meaningfully influences the optimization process.

(Empirical analysis) Following FalseReject, we evaluate the model's over-refusal rate by introducing a three-class metric, termed Useful Safety Rate (USR), which distinguishes among Direct Refusal, Safe Partial Compliance, and Full Compliance. In line with FalseReject, we employ the Claude Sonnet 4.6 model to classify the model outputs. We then compute the Useful Safety Rate (USR) using the following formulation; higher values indicate that the model is less prone to over-refusal caused by defensive handling of harmful samples. Formally:

$$\mathrm{USR}_{\mathrm{Benign}} = \frac{\#(\text{Full Compliance}) + \#(\text{Safe Partial Compliance})}{\#(\text{Total Benign Prompts})} \tag{1}$$

Please see the following results. All results are obtained using a Llama3-8B-Instruct model fine-tuned on GSM8K unless otherwise specified, and the fine-tuned model is evaluated on the test datasets provided by OKTest (300 samples) and FalseReject (500 samples).

(Test the model's over-refusal on the OKTest and FalseReject benchmarks.)

Methods	**Harmful Score ↓**	**Finetune Accuracy ↑**	**$\mathrm{USR}_{\text{Benign}}$ ↑**	**->**
Methods			OKTest	FalseReject
SFT	27.60	64.70	94.22	78.20
DSS	16.80	57.10	89.80	75.40
Surgery	8.90	68.50	89.00	72.80

The results above indicate that Surgery indeed increases the occurrence of Direct Refusal to some extent. Specifically, compared with SFT, $\mathrm{USR}_{\text{Benign}}$ decreases by 5.22% and 5.40% on OKTest and FalseReject, respectively; compared with DSS, which already has defensive capabilities, it decreases by 0.8% and 2.6%, respectively. This is mainly because the samples in these two datasets are highly similar to harmful samples, which can lead to misclassification. However, Surgery improves defensive performance by 18.7% and 7.9% compared with SFT and DSS, respectively. Therefore, **although Surgery introduces more Direct Refusal to some extent, the substantial improvement in overall defensive capability demonstrates that Surgery can effectively suppress harmful behaviors while maintaining an acceptable trade-off with over-refusal risk**.

(Summary) Surgery's over-refusal increases by approximately 5% compared with existing methods; however, its defensive performance improves by as much as 18.7%. Therefore, **Surgery's over-refusal remains within an acceptable range, while it achieves strong defensive performance.**

Comment 3.3: The experimental scale is somewhat limited. With only 1k training examples, full-parameter fine-tuning may lead to overfitting or catastrophic forgetting. A more practical setup should involve PEFT methods (e.g., LoRA) for experiments with 1k data points, while full-parameter fine-tuning would be more appropriate for larger datasets (e.g., 100k examples).

Response 3.3:

W3: The experimental scale is limited and needs to be evaluated under LoRA settings.

Please see the following results. All results are obtained using a Llama3-8B-Instruct model fine-tuned on GSM8K using LoRA with $r = 32$ and $\alpha = 4$, unless otherwise specified.

(Evaluation under LoRA settings)

Methods	Harmful Score ↓	Finetune Accuracy ↑
SFT	34.40	56.20
SPARD	40.40	59.10
DSS	19.80	63.00
Surgery	15.70	61.30

The results above show that **Surgery achieves better defensive performance and fine-tuning performance under the LoRA setting compared with the baselines**. Specifically, Surgery improves defensive performance by 18.7% and 5.1% over non-defensive SFT Harmful Score and Fine-tuning Accuracy, respectively. Furthermore, compared with existing defenses, it improves defensive performance by 24.7% and 4.1% over SPARD and DSS, respectively. This can be attributed to the fact that Surgery optimizes the model's intrinsic mechanisms.

(Summary) Surgery is applicable not only in full-parameter fine-tuning settings but also in parameter-efficient fine-tuning scenarios, such as LoRA.

Comment 3.4: Typo: line 206 Figure reffig:motivate ->Figure 3

Response 3.4:

Thank you for pointing this out. In the revision, we have corrected the typo in line 206 by updating "Figure reffig:motivate" to "Figure 3". In addition, we have carefully reviewed the entire manuscript to ensure no similar issues remain.

Comment 3.5 Q1: This paper mainly targets the FaaS scenario. However, in practice, when users attempt to fine-tune commercial LLMs (e.g., GPT-4o) through API-based services using harmful datasets, the process will be blocked by the service provider. A more realistic harmful tuning scenario is using only benign data to destroy the model's safety alignment [1–4]. It would be valuable to discuss whether Surgery remains effective under this setting.

Response 3.5:

Q1: It needs to be evaluated under benign data attack scenarios.

Thank you for the reviewer's valuable suggestion. In the revision, we evaluate our method under the four benign data attack scenarios [1-4] provided by the reviewer. Specifically, during fine-tuning, we incorporate high-intensity benign attack data selected from studies [1–4] into the fine-tuning process.

Please see the following results. All results are obtained using a Llama3-8B-Instruct model fine-tuned on GSM8K unless otherwise specified.

(Performance under different benign data attack settings)

	Harmful Score ↓	->	->	->	->	Finetune Accuracy ↑	->	->	->	->
Methods	clean	[1]	[2]	[3]	[4]	clean	[1]	[2]	[3]	[4]
SFT	15.10	17.30	19.10	16.80	14.90	67.20	64.40	67.80	65.50	64.50
DSS	13.50	13.60	16.10	14.60	14.80	59.80	67.60	55.20	57.90	51.60
SPARD	12.40	14.60	16.60	13.80	14.10	67.90	66.20	67.80	68.50	67.80
Surgery	5.50	6.80	5.90	5.40	8.10	69.40	67.40	67.80	69.80	69.00

(Observation 1) In the original version, we set the harmful data ratio p to 0 (i.e., clean) to simulate benign attacks. Compared with the baselines, Surgery achieves the best defensive and inference performance (i.e., 5.50% and 69.40%), indicating that Surgery can effectively mitigate the degradation in defensive performance caused by benign data.

(Observation 2) In this revision, we evaluate our method under the four benign data attack approaches provided by the reviewer (i.e., using selected benign data that can undermine the model's defensive capability during training). We observe that these selected data indeed degrade model performance; for example, under attack [2], SFT and DSS increase by 4% and 2.6%, respectively, compared with their clean counterparts. However, Surgery exhibits minimal fluctuation, with HS remaining around 6%, indicating that the attack has limited effectiveness against Surgery.

(Result) Surgery demonstrates strong defensive and inference performance in both randomly selected benign data training scenarios and targeted benign data attack scenarios.

Comment 3.6 Q2: The defense performance of Surgery relies heavily on the selection of anchor datasets (X_m and X_r). As noted in [5], harmful data from different distributions may exhibit substantially different characteristics. If the anchor data and the evaluation data come from different distributions, it remains unclear whether Surgery would still be effective. It would be helpful for the authors to discuss the robustness of the method under such distribution shifts, or provide empirical results evaluating this scenario.

Response 3.6:

Q2: It remains unclear whether Surgery is still effective when the anchor dataset and the evaluation dataset come from different distributions. The robustness of the method under distribution shift should be further discussed.

Thank you for the reviewer's valuable suggestion. In the original version, we used BeaverTails as the anchor dataset and evaluated poisoned harmful datasets from BeaverTails, Harmbench, and Sorrybench. In the revision, **we introduce an additional anchor dataset, RepNoise-Refusal**, and similarly evaluate performance on poisoned data from BeaverTails, Harmbench, and Sorrybench.

Please see the following results. All results are obtained using a Llama3-8B-Instruct model fine-tuned on GSM8K unless otherwise specified.

(Performance of Surgery under distribution shift)

Methods	Anchor Dataset	HS (BeaverTails) ↓	HS (Harmbench) ↓	HS (Sorrybench) ↓
SFT	—	27.60	28.75	30.91
Lisa	—	14.80	20.75	22.73
SafeGrad	—	18.50	23.50	23.64
ConstrainedSFT	—	17.60	21.25	28.41
AsFT	—	16.10	22.50	22.50
SPARD	—	20.60	30.00	27.73
DSS	—	16.80	24.50	25.68
Surgery (BeaverTails)	BeaverTails	8.90	9.50	12.95
Surgery (RepNoise-Refusal)	RepNoise-Refusal	13.20	12.40	16.36

(Observation) The above results demonstrate that **Surgery maintains reliable defensive performance even under distribution shift between the anchor datasets and harmful evaluation datasets.** Specifically, we observe that as the distribution gap between the anchor dataset and the harmful evaluation dataset increases, the defensive performance degrades. For example, when using RepNoise-Refusal as the anchor dataset, the defensive performance on BeaverTails drops by 4.3%. Nevertheless, compared with the baselines, Surgery still achieves the best defensive performance. *This further suggests that better-aligned (i.e., distribution-consistent) anchor datasets can lead to improved performance of Surgery.*

(Result) Distribution shift between the anchor datasets and harmful evaluation datasets affects the performance of Surgery to some extent, but it still maintains strong defensive capability and consistently outperforms the baselines.

Comment 3.7 Q3: The utility evaluation in this paper appears to be conducted in-distribution. For instance, on GSM8K, there is no significant performance difference between the fine-tuned model and the base model. It would be better to examine the method under a more out-of-distribution fine-tuning setting. For example, the authors could fine-tune the model on the full Alpaca dataset using Surgery, and then evaluate both the model's defense against harmful queries and its utility on GSM8K.

Response 3.7:

Q3: Can the method maintain both defensive performance and utility under out-of-distribution fine-tuning settings?

Thank you for the reviewer's valuable question. In the revised version, we conduct four groups of experiments: (1) training on GSM8K and evaluating on AGNEWS (**GSM8K_AGNEWS**); (2) training on AGNEWS and evaluating on GSM8K (**AGNEWS_GSM8K**); (3) training on SST2 and evaluating on AGNEWS (**SST2_AGNEWS**); and (4) training on AGNEWS and evaluating on SST2 (**AGNEWS_SST2**).

Please see the following results. All results are obtained using a Llama3-8B-Instruct model unless otherwise specified.

(Performance under out-of-distribution fine-tuning settings (GSM8K and AGNEWS))

	Harmful Score ↓	**->**	**Finetune Accuracy ↑**	**->**	**->**	**->**
Methods	GSM8K	AGNEWS	GSM8K_AGNEWS	AGNEWS_GSM8K	AGNEWS	GSM8K
SFT	27.60	31.00	72.10	74.60	90.30	64.70
Surgery	8.90	7.40	74.40	69.60	88.60	68.50

(Performance under out-of-distribution fine-tuning settings (SST2 and AGNEWS))

	Harmful Score ↓	**->**	**Finetune Accuracy ↑**	**->**	**->**	**->**
Methods	SST2	AGNEWS	SST2_AGNEWS	AGNEWS_SST2	SST2	AGNEWS
SFT	34.20	31.00	75.10	92.43	94.50	90.30
Surgery	9.40	7.40	74.40	91.86	94.50	88.60

(Observation) The above results show that under out-of-distribution fine-tuning settings, both SFT and Surgery suffer from degradation in fine-tuning performance. For example, GSM8K→AGNEWS is 18.20% and 14.20% lower than AGNEWS for SFT and Surgery, respectively; similarly, AGNEWS→SST2 is 2.07% and 2.64% lower than SST2 for SFT and Surgery, respectively. We observe that the performance degradation of SFT and Surgery is comparable, suggesting that out-of-distribution fine-tuning inherently leads to model forgetting. **Importantly, compared with standard SFT, Surgery does not introduce additional performance loss.**

(Result) Out-of-distribution fine-tuning inherently leads to model forgetting when using SFT. However, compared with standard SFT, Surgery does not introduce additional performance loss.